

WEEK – 1

DATA CLEANING AND TRANSFORMATION REPORT

Introduction

Power BI is a tool used to work with data and create reports and dashboards. In this task, I used Power BI to import and clean the Superstore Sales Dataset using Power Query Editor. I checked the data types, missing values, record uniqueness, and text formatting. I also performed column transformations such as renaming and splitting columns. From the Order Date column, I created Year, Quarter, Month, Day, and Day of Week information. Finally, I validated the cleaned data and applied the changes in Power BI.

Objectives

- * To import the Superstore Sales Dataset into Power BI.
- * To navigate Report, Data, and Model views.
- * To inspect the dataset using Power Query Editor.
- * To check and correct the data types.
- * To check missing and null values.
- * To check record uniqueness.
- * To standardize text formatting using Trim and Clean.
- * To perform column transformations such as Rename and Split.
- * To extract Year, Quarter, Month, Day, and Day of Week from Order Date.
- * To load the cleaned data into the Power BI data model.

Tools and Dataset

Tool Used: Microsoft Power BI Desktop

Data Preparation Tool: Power Query Editor

Dataset Name: Superstore Sales Dataset

File Type: CSV

Power Query Editor was used to import, inspect, clean, and transform the dataset before loading it into Power BI.

1. Importing the Dataset

First, I opened Power BI Desktop and imported the Superstore Sales Dataset. After importing the file, I opened it in Power Query Editor for data cleaning and transformation.

Steps Followed

1. Opened Power BI Desktop.
2. Clicked *Home*.
3. Selected *Get Data*.
4. Selected *Text/CSV*.
5. Selected the Superstore Sales Dataset.
6. Clicked *Open*.
7. Checked the data preview.
8. Selected *Transform Data*.
9. The dataset opened in Power Query Editor.

samplesuperstore.csv

File Origin

65001: Unicode (UTF-8)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelp
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
8	US-2023-106054	1/6/2023	1/7/2023	First Class	JO-15145	Jack O'Briant	Corporate	United States	Athens
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
12	US-2023-130813	1/6/2023	1/8/2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Angel
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderso
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville
17	US-2023-135405	1/9/2023	1/13/2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo
18	US-2023-135405	1/9/2023	1/13/2023	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo
19	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfiel
20	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfiel

Extract Table Using Examples

Load

Transform Data

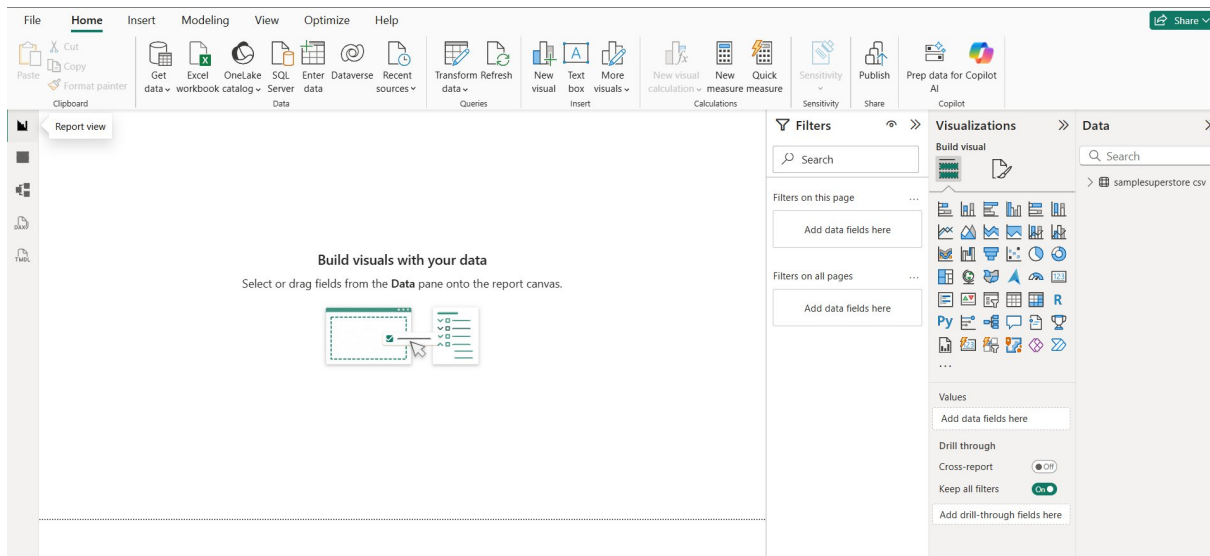
Cancel

2.Navigating Power BI Views

Power BI provides three main views for working with data.

Report View

Report View is used to create visualizations, charts, and reports.



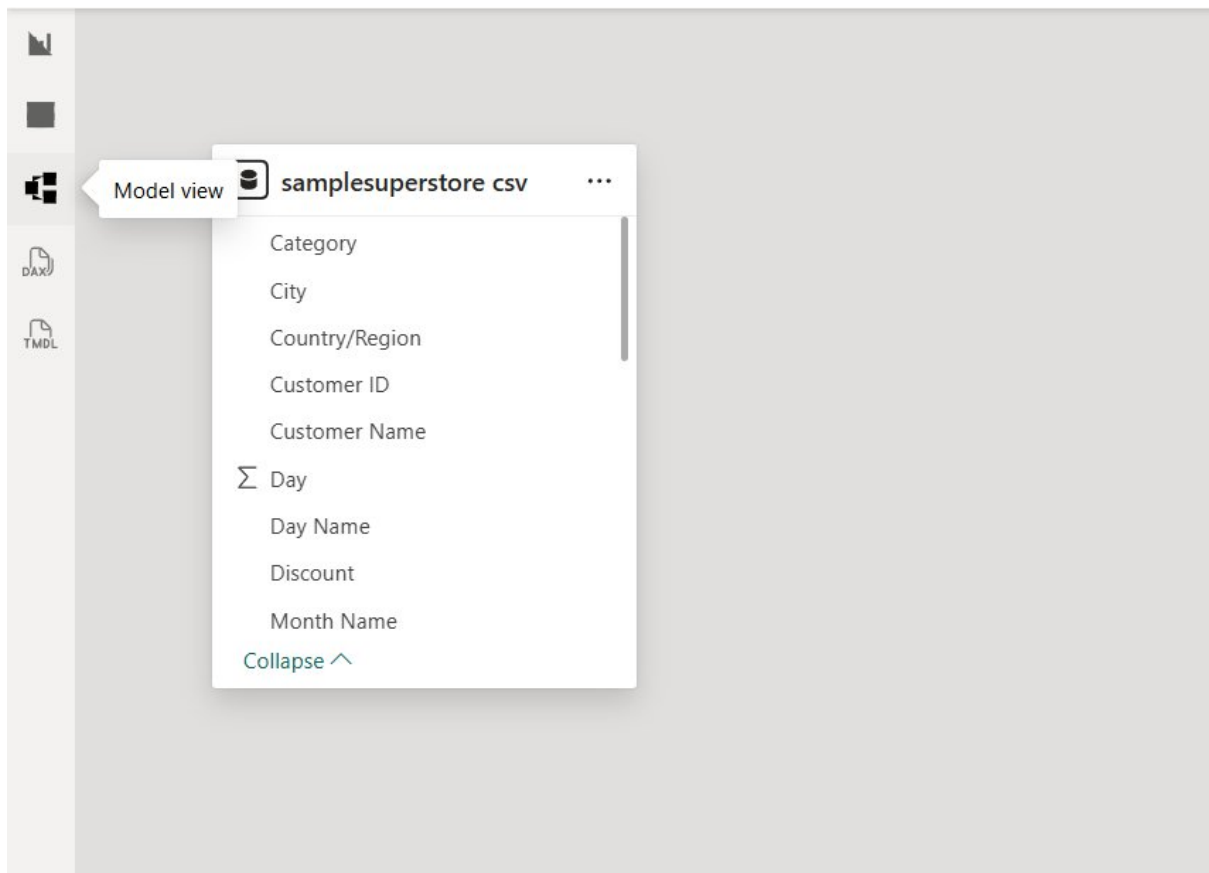
Data View

Data View is used to inspect the rows and columns of the loaded data.

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	Postal Code	Region
1	S-2023-130813	06 January 2023	08 January 2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Angeles	California	90049	V
57	US-2023-148614	20 January 2023	25 January 2023	Standard Class	MV-17485	Mark Van Huff	Consumer	United States	Los Angeles	California	90049	V
122	US-2023-164903	20 February 2023	24 February 2023	Standard Class	SR-20740	Steven Roelle	Home Office	United States	Los Angeles	California	90049	V
331	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	V
332	US-2023-102652	06 April 2023	12 April 2023	Standard Class	AY-10555	Andy Yotov	Corporate	United States	Los Angeles	California	90049	V
370	US-2023-120740	15 April 2023	15 April 2023	Same Day	PS-18970	Paul Stevenson	Home Office	United States	Los Angeles	California	90049	V
373	US-2023-110849	18 April 2023	23 April 2023	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
375	US-2023-110849	18 April 2023	23 April 2023	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
377	US-2023-110849	18 April 2023	23 April 2023	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	V
723	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	V
727	US-2023-114125	09 July 2023	13 July 2023	Standard Class	GH-14410	Gary Hansen	Home Office	United States	Los Angeles	California	90049	V
1409	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1410	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1411	US-2023-101931	28 October 2023	31 October 2023	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	V
1422	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1427	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1428	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1430	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1431	US-2023-106439	31 October 2023	04 November 2023	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	V
1446	US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	V
1449	US-2023-168305	01 November 2023	08 November 2023	Standard Class	PL-18925	Paul Lucas	Home Office	United States	Los Angeles	California	90049	V
1468	US-2023-153619	03 November 2023	03 November 2023	Same Day	TC-21535	Tracy Collins	Home Office	United States	Los Angeles	California	90049	V
1816	US-2023-138737	07 December 2023	10 December 2023	First Class	FP-14320	Frank Preis	Consumer	United States	Los Angeles	California	90049	V
1859	US-2023-113579	13 December 2023	15 December 2023	Second Class	KD-16345	Katherine Ducich	Consumer	United States	Los Angeles	California	90049	V
2288	US-2024-138219	23 March 2024	30 March 2024	Standard Class	BP-11095	Bart Pistole	Corporate	United States	Los Angeles	California	90049	V
3056	US-2024-121188	28 August 2024	04 September 2024	Standard Class	CB-12025	Cassandra Brandow	Consumer	United States	Los Angeles	California	90049	V

Model View

Model View is used to view and manage tables and relationships in the data model.



Steps Followed

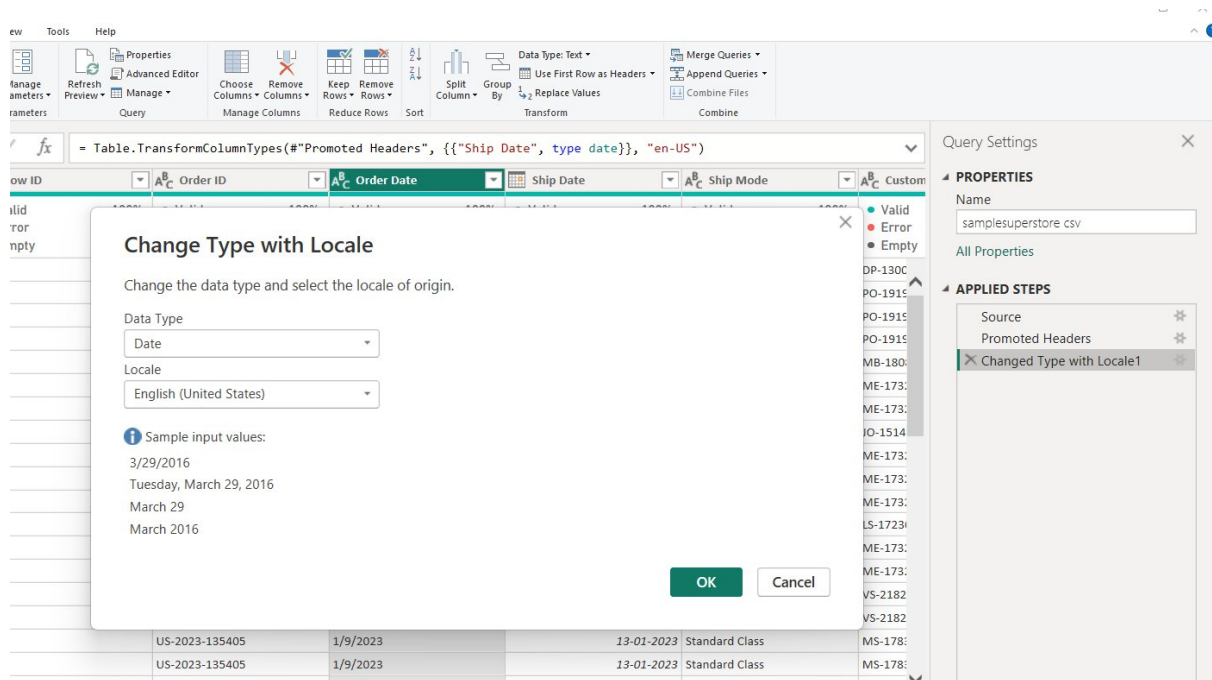
1. Opened Power BI Desktop.
2. Checked the *Report View*.
3. Checked the *Data View*.
4. Checked the *Model View*.
5. Reviewed the available dataset and data model.

3. Checking and Changing Data Types

After importing the dataset, I checked the data type of each required column. Correct data types are important for accurate data processing and analysis.

Steps Followed

1. Opened the dataset in Power Query Editor.
2. Selected the column that I wanted to check.
3. Checked the data type shown near the column name.
4. If the data type was incorrect, selected the column.
5. Selected *Data Type / Change Type*.
6. Selected the required data type.
7. Set text columns to *Text*.
8. Set quantity-related columns to *Whole Number*.
9. Set sales and profit-related columns to *Decimal Number*.
10. Set *Order Date* to *Date*.
11. Repeated the process for other required columns.



4. Cleaning Text Data Using Trim and Clean

After checking the data types, I cleaned the categorical text data. The required categorical attributes were *Category, Sub-Category, and Segment*.

Steps Followed

1. Selected the *Category* column.
2. Selected *Transform*.
3. Selected *Format*.
4. Selected *Trim*.
5. Checked the cleaned values.
6. Selected *Transform → Format → Clean*.
7. Checked the data again.
8. Repeated the same process for *Sub-Category*.
9. Repeated the same process for *Segment*.

Trim was used to remove unnecessary spaces.

The screenshot shows the Power Query Editor with a table of product data. The 'Product Name' column is selected, and a context menu is open with the 'Trim' option highlighted. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, where 'Trimmed Text1' is the current step. The table has columns: Product ID, Category, Sub-Category, Product Name, and Sales. The 'Product Name' column header is highlighted in blue.

Clean was used to remove unwanted non-printable characters.

The screenshot shows the Power Query Editor with the same table. The 'Product Name' column is selected, and a context menu is open with the 'Clean' option highlighted. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, where 'Cleaned Text1' is the current step. The table has columns: Product ID, Category, Sub-Category, Product Name, and Sales. The 'Product Name' column header is highlighted in blue.

5. Checking Blank, Empty, Null and Error Values

After cleaning the text data, I checked the dataset for blank, empty, null, and error values.

Steps Followed

1. Selected the required column.
2. Clicked the *Filter* option.
3. Checked the values shown in the filter list.
4. Looked for blank or empty values.
5. Checked for null values.
6. Checked for Error values.
7. Reviewed the affected columns.
8. Handled unnecessary values where required.
9. Rechecked the columns after cleaning.

= Table.TransformColumns("#Cleaned Text1",{{"Product Name", Text.Trim, type text}})			
	Product Name	1.2 Sales	
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
1	Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets...		
2	Gbc Standard Plastic Binding Systems Combs		
3	Avery 508		
4	Safco Boltless Steel Shelving		
5	Avery Hi-Liter Everbold Pen Style Fluorescent Highlighters, 4/Pack		
6	Global Deluxe High-Back Manager'S Chair		
7	Rogers Handheld Barrel Pencil Sharpener		
8	Dixon Prang Watercolor Pencils, 10-Color Set With Brush		
9	Ibico Hi-Tech Manual Binding System		
10	Alliance Super-Size Bands, Assorted Sizes		
11	Southworth 25% Cotton Granite Paper & Envelopes		
12	Xerox 225		
13	Wireless Extenders Zboost Yx545 Soho Signal Booster		
14	Ge 30524Ee4		
15	Howard Miller 14-1/2" Diameter Chrome Round Wall Clock		
16	Acco Four Pocket Poly Ring Binder With Label Holder, Smoke, 1"	10.43	7
17	Newell 312	9.344	2
18	Memorex Micro Travel Drive 8 Gb	31.2	3
19	Howard Miller 11-1/2" Diameter Ridgewood Wall Clock	51.94	1
20	Avery 482	2.89	1
21	Dax Value U-Channel Document Frames, Easel Back	9.94	2
22	O'Sullivan Elevations Bookcase, Cherry Finish	333.999	3
23	Global Highback Leather Tilter In Burgundy	545.94	6
24	4009 Highlighters By Sanford	19.9	5
25	Avery Metallic Poly Binders	3.438	2
26	2 Ring Staple Book	5.54	2

6.Performing Column Transformations

The dataset was transformed according to the requirements of the task.

Rename Column

1. Selected the required column.
2. Right-clicked the column.
3. Selected *Rename*.
4. Entered the required column name.
5. Checked the renamed column.

The screenshot displays a Power BI Desktop interface. The main area shows a data table with columns: Discount, Profit, order year, Month Name, Day, and Day Name. A right-click context menu is open over the 'order year' column, with the 'Rename...' option highlighted. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Renamed Columns' at the bottom.

	Discount	Profit	order year	Month Name	Day	Day Name
1	2 0.2	5.5512			3	Tuesday
2	2 0.8	-5.487			4	Wednesday
3	3 0.2	4.2717			4	Wednesday
4	3 0.2	-64.7748			4	Wednesday
5	3 0.2	4.884			5	Thursday
6	9 0	746.4078			6	Friday
7	2 0	1.4796			6	Friday
8	3 0	5.2398			6	Friday
9	2 0	274.491			6	Friday
10	4 0	0.3112			6	Friday
11	1 0	3.0084			6	Friday
12	3 0	9.3312			6	Friday
13	4 0	204.1092			6	Friday
14	2 0	113.6742			6	Friday
15	3 0.6	-53.7096			7	Saturday
16	7 0.8	-18.2525			7	Saturday
17	2 0.2	1.168			9	Monday
18	3 0.2	9.75			9	Monday
19	1 0	21.2954			10	Tuesday
20	1 0	1.3583			10	Tuesday
21	2 0	3.0814			11	Wednesday
22	3 0.15	3.9294			13	Friday
23	6 0	87.3504			13	Friday
24	5 0	6.567			13	Friday
25	2 0.7	-2.5212			13	Friday
26						

7.Creating Date Attributes from Order Date

After setting *Order Date* to the Date data type, I extracted useful temporal attributes from it.

The required attributes are:

- * Year
- * Quarter
- * Month
- * Day
- * Day of Week

A. Creating Year

1. Selected the *Order Date* column.
2. Went to *Add Column*.
3. Selected *Date*.
4. Selected *Year*.
5. Selected *Year*.
6. Checked the new Year column.

The screenshot displays the Power BI Desktop interface. The main data table, 'samplesuperstore...', is visible with columns: Row ID, Order ID, Order Date, Ship Mode, Customer ID, and Customer Name. The 'Order Date' column is selected. The 'Add Column' menu is open, showing the path: Date > Year > Year. The 'Year' column has been added to the table. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Inserted Year' as the most recent step.

Row ID	Order ID	Order Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	05-01-2023	Standard Class	DP-13000	Darren Power
2	US-2023-112326	08-01-2023	Standard Class	PO-19195	Phyllina Ober
3	US-2023-112326	08-01-2023	Standard Class	PO-19195	Phyllina Ober
4	US-2023-112326	08-01-2023	Standard Class	PO-19195	Phyllina Ober
5	US-2023-141817	12-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
7	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
8	US-2023-106054	07-01-2023	First Class	JO-15145	Jack O'Briant
9	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
10	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
11	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
12	US-2023-130813	08-01-2023	Second Class	LS-17230	Lycoris Saund
13	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
14	US-2023-167199	10-01-2023	Standard Class	ME-17320	Maria Etezadi
15	US-2023-105417	12-01-2023	Standard Class	VS-21820	Vivek Sundar
16	US-2023-105417	12-01-2023	Standard Class	VS-21820	Vivek Sundar
17	US-2023-135405	13-01-2023	Standard Class	MS-17830	Melanie Seite
18	US-2023-135405	13-01-2023	Standard Class	MS-17830	Melanie Seite
19	US-2023-149020	15-01-2023	Standard Class	AI-10780	Anthony Jaco
20	US-2023-149020	15-01-2023	Standard Class	AI-10780	Anthony Jaco
21	US-2023-130092	14-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	18-01-2023	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	16-01-2023	Second Class	ND-18370	Natalie DeCh
24	US-2023-157147	18-01-2023	Standard Class	BD-11605	Brian Dahlen
25	US-2023-118192	18-01-2023	Standard Class	MM-17920	Michael Moon
26	US-2023-163326	15-01-2023	Second Class	CE-13320	Chae Collette

B. Creating Quarter

1. Selected the *Order Date* column.
2. Went to *Add Column*.
3. Selected *Date*.
4. Selected *Quarter*.
5. Selected *Quarter of Year*.
6. Checked the new Quarter column.

The screenshot displays the Power Query Editor interface. The 'Add Column' menu is open, showing the path: Date > Quarter > Quarter of Year. The main table view shows columns for Row ID, Order ID, Order Date, Ship Mode, Customer ID, and Customer Name. The 'Order Date' column has a data type of 'Date'. The 'Quarter of Year' column is being added, with a formula bar showing 'Table.AddColumn(#\"Changed Type with Locale\", \"Quarter of Year\", Int64.Type)'. The 'APPLIED STEPS' pane on the right shows the sequence of transformations applied to the data.

C. Creating Month

1. Selected the *Order Date* column.
2. Went to *Add Column*.
3. Selected *Date*.
4. Selected *Month*.
5. Selected the required month option.
6. Checked the new Month column.

Queries [1] fx Table.AddColumn(*Changed Type with Locale*, "Year", Int64.Type)

Row ID	Order ID	Order Date
1	US-2023-103800	
2	US-2023-112326	
3	US-2023-112326	
4	US-2023-112326	
5	US-2023-141817	05-01-2023
6	US-2023-167199	06-01-2023
7	US-2023-167199	06-01-2023
8	US-2023-106054	06-01-2023
9	US-2023-167199	06-01-2023
10	US-2023-167199	06-01-2023
11	US-2023-167199	06-01-2023
12	US-2023-130813	06-01-2023
13	US-2023-167199	06-01-2023
14	US-2023-167199	06-01-2023
15	US-2023-105417	07-01-2023
16	US-2023-105417	07-01-2023
17	US-2023-135405	09-01-2023
18	US-2023-135405	09-01-2023
19	US-2023-149020	10-01-2023
20	US-2023-149020	10-01-2023
21	US-2023-130092	11-01-2023
22	US-2023-157147	13-01-2023
23	US-2023-109232	13-01-2023
24	US-2023-157147	13-01-2023
25	US-2023-118192	13-01-2023

Query Settings

PROPERTIES

Name: samplesuperstore.csv

APPLIED STEPS

Source

Promoted Headers

Changed Type with Locale1

Trimmed Text

Cleaned Text

Changed Type

Removed Errors

Capitalized Each Word

Cleaned Text1

Trimmed Text1

Changed Type with Locale

Inserted Year

Inserted Month Name

Inserted Day

Inserted Day Name

Renamed Columns

Split Column by Delimiter

Changed Type1

D. Creating Day

1. Selected the *Order Date* column.
2. Went to *Add Column*.
3. Selected *Date*.
4. Selected *Day*.
5. Selected *Day*.
6. Checked the new Day column.

Queries [1] fx Table.AddColumn(*Changed Type with Locale*, "Day", Int64.Type)

Row ID	Order ID	Order Date	Day
1	US-2023-103800		
2	US-2023-112326		
3	US-2023-112326		
4	US-2023-112326		
5	US-2023-141817	05-01-2023	5
6	US-2023-167199	06-01-2023	6
7	US-2023-167199	06-01-2023	6
8	US-2023-106054	06-01-2023	6
9	US-2023-167199	06-01-2023	6
10	US-2023-167199	06-01-2023	6
11	US-2023-167199	06-01-2023	6
12	US-2023-130813	06-01-2023	6
13	US-2023-167199	06-01-2023	6
14	US-2023-167199	06-01-2023	6
15	US-2023-105417	07-01-2023	7
16	US-2023-105417	07-01-2023	7
17	US-2023-135405	09-01-2023	9
18	US-2023-135405	09-01-2023	9
19	US-2023-149020	10-01-2023	10
20	US-2023-149020	10-01-2023	10
21	US-2023-130092	11-01-2023	11
22	US-2023-157147	13-01-2023	13
23	US-2023-109232	13-01-2023	13
24	US-2023-157147	13-01-2023	13
25	US-2023-118192	13-01-2023	13

Query Settings

PROPERTIES

Name: samplesuperstore.csv

APPLIED STEPS

Source

Promoted Headers

Changed Type with Locale1

Trimmed Text

Cleaned Text

Changed Type

Removed Errors

Capitalized Each Word

Cleaned Text1

Trimmed Text1

Changed Type with Locale

Inserted Year

Inserted Month Name

Inserted Day

Inserted Day Name

Renamed Columns

Split Column by Delimiter

Changed Type1

E. Creating Day of Week

1. Selected the *Order Date* column.
2. Went to *Add Column*.
3. Selected *Date*.
4. Selected *Day*.
5. Selected *Name of Day*.
6. Checked the new Day of Week column.

The screenshot shows the Power BI Desktop interface. The 'Add Column' menu is open for the 'Order Date' column. The 'Name of Day' option is selected. The background table shows data for 'samplesuperstore.csv' with columns: Row ID, Order ID, Order Date, Ship Mode, Customer ID, and Customer Name. The 'Order Date' column has a validation status of 100% Valid. The 'Ship Mode' column has a validation status of 100% Valid. The 'Customer ID' column has a validation status of 100% Valid. The 'Customer Name' column has a validation status of 100% Valid. The 'Name of Day' column is being added to the table.

These date attributes can be used for further time-based analysis in Power BI.

8.Final Data Validation

After completing the cleaning and transformation steps, I checked the dataset one more time.

Final Checks

- * Checked the data types.
- * Checked the Order Date column.
- * Checked Year, Quarter, Month, Day, and Day of Week columns.
- * Checked blank and empty values.
- * Checked null values.
- * Checked error values.
- * Checked the cleaned text data.
- * Checked the renamed columns.
- * Checked the transformed columns.
- * Reviewed the final dataset.

The screenshot shows a data transformation tool interface. The main table displays data with columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. Each column has a data quality summary (Valid, Error, Empty) and a percentage. The 'Order Date' column is highlighted. To the right, a 'Query Settings' panel shows the 'APPLIED STEPS' list, which includes: Source, Promoted Headers, Changed Type with Locale1, Trimmed Text, Cleaned Text, Changed Type, Removed Errors, Capitalized Each Word, Cleaned Text1, Trimmed Text1, Changed Type with Locale, Inserted Year, Inserted Month Name, Inserted Day, Inserted Day Name, Renamed Columns, Split Column by Delimiter, and Changed Type1.

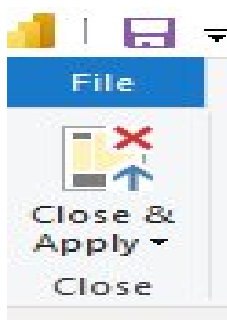
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Power
2	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
3	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
4	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phillina Ober
5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
8	US-2023-106054	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Briant
9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
12	US-2023-130813	06-01-2023	08-01-2023	Second Class	LS-17230	Lycoris Saund
13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezadi
15	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundar
16	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundar
17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
21	US-2023-130092	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlien
23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18370	Natalie DeChi
24	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlien
25	US-2023-118192	13-01-2023	18-01-2023	Standard Class	MM-17920	Michael Moo
26	US-2023-163226	13-01-2023	15-01-2023	Standard Class	CE-13060	Cheryl Evans

9.Close and Apply

After completing the cleaning and transformation process, I applied the changes to Power BI.

Steps Followed

1. Checked the final dataset in Power Query Editor.
2. Went to the *Home* tab.
3. Clicked *Close & Apply*.
4. Waited for Power BI to apply the changes.
5. The cleaned dataset was loaded into the Power BI data model.



Final Result

The Superstore Sales Dataset was successfully imported, cleaned, and transformed using Power BI and Power Query Editor. The data types were checked and corrected, and the categorical text data was standardized using Trim and Clean. Blank, empty, null, and error values were checked. Column transformations such as Rename and Split were performed where required. The Order Date column was used to create Year, Quarter, Month, Day, and Day of

Week attributes. Finally, the cleaned dataset was validated and loaded into the Power BI data model.

Conclusion

In this task, I successfully learned how to import, clean, and transform the Superstore Sales Dataset using Microsoft Power BI and Power Query Editor. I learned how to check data types, missing values, record uniqueness, and text formatting. I also performed column transformations and extracted useful temporal information from the Order Date column. Finally, I validated the cleaned dataset and used Close & Apply to load the transformations into Power BI. This task helped me understand the basic data preparation process required before creating reports and dashboards.